

Example 1: The following matrices are in echelon form. The leading entries ($\square$) may have any nonzero value; the starred entries (*) may have any value (including zero).

$$\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & \square & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \square & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \square & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \square & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \square & * \end{bmatrix}$$

The following matrices are in reduced echelon form because the leading entries are 1's, and there are 0's below and above each leading 1.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Theorem 1: Each matrix is row equivalent to one and only one reduced echelon matrix.

The proof is given in Appendix A and one must study Chapter 4 first before understanding it.

Pivot Positions

When row operations on a matrix produce an echelon form, further row operations to obtain the reduced echelon form do not change the positions of the leading entries. Since the reduced echelon form is unique, the leading entries are always in the same positions in any echelon form obtained from a given matrix. These leading entries correspond to leading 1's in the reduced echelon form.

Definition: A pivot position in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A pivot column is a column of A that contains a pivot position.